Virtual Element Method for oral biofilm mechanics: from confocal images to viscoelastic fracture on arbitrary polygonal meshes

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Abstract

Oral biofilms are structured multispecies communities whose mechanical properties depend on microbial composition. We present a Virtual Element Method (VEM) framework that reduces the image-to-mechanics pipeline from five steps (confocal segmentation, voxelization, marching cubes, tetrahedral meshing, FEM) to two (Voronoi tessellation from colony detection, VEM solve). The framework comprises 12 solver modules—linear and nonlinear elasticity, Standard Linear Solid (SLS) viscoelasticity with the Simo (1987) exponential integrator, phase-field fracture, cohesive zone modelling, and adaptive h-refinement—all operating on arbitrary polygonal or polyhedral elements. Material properties are assigned per colony via a Dysbiosis Index (DI) derived from TMCMC-calibrated species dynamics, yielding DI-dependent stiffness $E(\mathrm{DI})$, fracture toughness $G_c(\mathrm{DI})$, and SLS relaxation parameters. The viscoelastic VEM achieves machine-precision agreement with analytical solutions in both 2D (1.3×10^{-15} relative error) and 3D (4.9×10^{-16}). Growth-coupled simulations reveal divergent mechanical evolution: commensal biofilms stiffen (E_∞: 481→913 Pa) while dysbiotic biofilms soften (E_∞: 481→324 Pa) under identical initial conditions. To our knowledge, this is the first application of VEM to biofilm mechanics.

Keywords: Virtual Element Method, biofilm mechanics, viscoelasticity, phase-field fracture, Dysbiosis Index, confocal imaging

1 Introduction

Oral biofilms are structured multispecies communities that colonize tooth surfaces, forming a mechanically significant interface between the tooth substrate and the oral cavity [13]. The transition from a healthy (commensal) to a disease-associated (dysbiotic) state is driven by shifts in species composition [17], which in turn affect the biofilm's elastic modulus [10, 11], viscoelastic relaxation behaviour [12], and fracture toughness [14, 15]. Understanding these composition–mechanics links is essential for predicting biofilm detachment, a key event in both periodontitis progression and therapeutic intervention.

Current computational approaches to biofilm mechanics rely on the finite element method (FEM) with structured meshes. Klempt et al. [16] demonstrated staggered coupling of species dynamics with continuum mechanics, but the mesh generation pipeline (confocal image → voxel segmentation → marching cubes → tetrahedral mesh → Abaqus input) involves five labor-intensive steps. Moreover, standard FEM mesh generators cannot naturally accommodate the arbitrary colony shapes observed in confocal images, requiring substantial manual effort for mesh-to-colony correspondence.

The Virtual Element Method (VEM) [1, 2] offers a compelling alternative. As a generalisation of FEM, VEM solves partial differential equations on meshes of *arbitrary polygonal or polyhedral elements*,

eliminating the need for standard element topologies. The key innovation is the *elliptic projection* Π^∇ that maps *virtual* (implicitly defined) basis functions onto a polynomial subspace, from which the stiffness matrix is assembled without ever constructing explicit shape functions [3].

The VEM has been extensively developed for solid mechanics at IKM Hannover: Wriggers and Hudobivnik [4] formulated low-order 3D VEM for elastoplasticity, Aldakheel et al. [6] introduced phase-field brittle fracture on VEM meshes, and the comprehensive monograph [5] covers applications from linear elasticity to contact mechanics.

For biofilm mechanics, VEM offers three specific advantages:

1. **Mesh–image compatibility**: Voronoi tessellation from colony centroids yields a mesh where each cell corresponds 1:1 to a micro-colony, enabling a 2-step pipeline (Fig. 1).

2. **Per-colony material assignment**: Each element receives its own DI-dependent material parameters without sub-element averaging.

3. **Topology tolerance**: Growth, detachment, and crack propagation change the domain topology; VEM handles arbitrary element shapes without global remeshing.

This paper presents a comprehensive VEM framework for biofilm mechanics comprising 12 solver modules and over 120 automated tests. We focus on three key contributions: (i) a viscoelastic VEM with the Simo [7] exponential integrator, validated to machine precision in 2D and 3D; (ii) DI-dependent constitutive laws for stiffness, fracture toughness, and viscoelastic relaxation; (iii) a growth-coupled VE-VEM that reveals divergent mechanical evolution of commensal vs. dysbiotic biofilms.

2 Mathematical framework

2.1 VEM for linear elasticity

Consider a domain $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) decomposed into a mesh $\mathcal{T}_h$ of non-overlapping polygonal (d=2) or polyhedral (d=3) elements E. For the lowest-order case (k=1), the local virtual element space on element E with n_v vertices is

$$V_h^E = \left\{ \boldsymbol{v}_h \in [H^1(E)]^d \;:\; \boldsymbol{v}_h|_{\partial E} \text{ piecewise linear}, \; \Delta \boldsymbol{v}_h = \mathbf{0} \text{ in } E \right\}, \tag{1}$$

with degrees of freedom $\{\boldsymbol{u}_i\}_{i=1}^{n_v}$ at the vertices, giving $n_{\text{dof}}^E = d \cdot n_v$.

The elliptic projection $\Pi^\nabla : V_h^E \to [\mathcal{P}_k(E)]^d$ minimises $a^E(\boldsymbol{v}_h - \Pi^\nabla \boldsymbol{v}_h, \boldsymbol{p})$ for all $\boldsymbol{p} \in [\mathcal{P}_k]^d$, where $a^E(\cdot,\cdot)$ is the element bilinear form. For k=1 in 2D, the polynomial space has 6 modes (3 rigid-body + 3 constant strain); in 3D it has 12 modes (3 translations + 3 rotations + 6 strain).

The projection is computed from the matrices $\mathbf{G} = \mathbf{B}\,\mathbf{D}$ and $\Pi^\nabla = \mathbf{G}^{-1}\,\mathbf{B}$, where $\mathbf{D}$ evaluates polynomials at vertex DOFs and $\mathbf{B}$ encodes the bilinear form via edge (2D) or face (3D) integrals [3]. The element stiffness matrix decomposes as

$$\mathbf{K}^E = \mathbf{K}_\pi^E + \mathbf{K}_{\text{stab}}^E, \tag{2}$$

where $\mathbf{K}_\pi^E = (\Pi^\nabla)^\top \tilde{\mathbf{C}}\,\Pi^\nabla$ provides consistency and $\mathbf{K}_{\text{stab}}^E = \alpha_s \operatorname{tr}(\mathbf{C})\,|E|\,(\mathbf{I} - \Pi^\nabla \mathbf{D})^\top (\mathbf{I} - \Pi^\nabla \mathbf{D})$ provides stability with $\alpha_s = 0.5$.

2.2 Viscoelastic VEM with the Simo integrator

We adopt the Standard Linear Solid (SLS / Zener) model with DI-dependent parameters (Section 3):

$$\boldsymbol{\sigma}(t) = \mathbf{C}_\infty \boldsymbol{\varepsilon}(t) + \boldsymbol{h}(t), \tag{3}$$

where $\mathbf{C}_\infty$ and $\mathbf{C}_1 = \mathbf{C}_0 - \mathbf{C}_\infty$ are the equilibrium and overstress constitutive matrices, and $\boldsymbol{h}$ is the internal viscous stress satisfying

$$\dot{\boldsymbol{h}} = \mathbf{C}_1\,\dot{\boldsymbol{\varepsilon}} - \boldsymbol{h}/\tau. \tag{4}$$

Following Simo [7], Eq. (4) is integrated exactly over a time step $\Delta t = t_{n+1} - t_n$:

$$\boldsymbol{h}_{n+1} = e^{-\Delta t/\tau}\,\boldsymbol{h}_n + \gamma\,\mathbf{C}_1\,\Delta\boldsymbol{\varepsilon}, \qquad \gamma = \frac{\tau}{\Delta t}\big(1 - e^{-\Delta t/\tau}\big). \tag{5}$$

The algorithmic tangent modulus is $\mathbf{C}_{\text{alg}} = \mathbf{C}_\infty + \gamma\,\mathbf{C}_1$, which enters the VEM stiffness as

$$\mathbf{K}_{\text{alg}}^E = (\boldsymbol{\Pi}_\varepsilon^E)^\top\,\mathbf{C}_{\text{alg}}^E\,|E|\,\boldsymbol{\Pi}_\varepsilon^E + \mathbf{K}_{\text{stab}}^E, \tag{6}$$

where $\mathbf{\Pi}_\varepsilon^E$ is the strain projector ($3 \times 2n_v$ in 2D, $6 \times 3n_v$ in 3D). At each time step, the right-hand side includes the viscous pre-stress contribution $\boldsymbol{f}_h^E = -(\mathbf{\Pi}_\varepsilon^E)^\top |E|\, e^{-\Delta t/\tau}\, \boldsymbol{h}_n^E$.

This formulation is *unconditionally stable* for any $\Delta t > 0$ and exact for constant strain rate within a time step.

2.3 Phase-field fracture on VEM

Following Aldakheel et al. [6], the regularised crack topology is described by a phase-field $d \in [0,1]$ (d=0: intact, d=1: fully cracked). The total energy functional reads

$$\mathcal{E}(\boldsymbol{u}, d) = \int_\Omega g(d)\, \psi^+(\boldsymbol{\varepsilon}) + \psi^-(\boldsymbol{\varepsilon})\, \mathrm{d}\Omega + \int_\Omega G_c \Big(\frac{d^2}{2\ell} + \frac{\ell}{2} |\nabla d|^2 \Big)\, \mathrm{d}\Omega, \tag{7}$$

where $g(d) = (1-d)^2 + \eta_{\mathrm{res}}$ is the degradation function, ℓ is the length scale, and $\psi^\pm$ is the spectral decomposition [8]. The staggered solution alternates between the displacement and phase-field subproblems until convergence. The fracture toughness $G_c(\mathrm{DI})$ is DI-dependent (Section 3.3), making dysbiotic regions inherently more fragile.

2.4 Additional modules

The framework includes several further capabilities:

- **Neo-Hookean VEM**: Large-deformation hyperelasticity with $W(\mathbf{F}) = \frac{\mu}{2}(I_1 - d) - \mu \ln J + \frac{\lambda}{2}(\ln J)^2$, solved via Newton-Raphson with load stepping and line search.
- **Cohesive zone model (CZM)**: Bilinear traction-separation law [9] with DI-dependent cohesive strength $\sigma_{\max(\mathrm{DI})}$ for the tooth–biofilm interface.
- **P_2 VEM**: Second-order elements with vertex + edge-midpoint DOFs, achieving 15–45% stress accuracy improvement over P_1.
- **Adaptive h-refinement**: A posteriori error estimator with Dörfler marking for automatic mesh refinement at crack tips.

3 DI-dependent constitutive laws

3.1 Dysbiosis Index

The Dysbiosis Index is defined as the normalised Shannon entropy of the species fractions ϕ_i ($i = 1, \ldots, 5$):

$$\mathrm{DI} = -\frac{1}{\ln 5} \sum_{i=1}^{5} \phi_i \ln \phi_i, \qquad \mathrm{DI} \in [0,1]. \tag{8}$$

Species fractions are obtained from a Hamilton 5-species replicator ODE calibrated by Transitional Markov Chain Monte Carlo (TMCMC) inference with 20 parameters across 4 experimental conditions [16, 17].

3.2 Stiffness: $E(\mathrm{DI})$

$$E(\mathrm{DI}) = E_{\min} + (E_{\max} - E_{\min})(1-\mathrm{DI})^n, \tag{9}$$

with $E_{\max=1000}\,\mathrm{Pa}$, $E_{\min=30}\,\mathrm{Pa}$, $n = 2$. The 30-fold stiffness range and power-law exponent are supported by AFM measurements on oral biofilms with simultaneous 16S rRNA species quantification [10, 11] and percolation theory ($n \approx 2$ for 3D elastic networks near the rigidity threshold [18, 19]).

3.3 Fracture toughness: $G_c(\mathrm{DI})$

$$G_c(\mathrm{DI}) = G_{c,\min} + (G_{c,\max} - G_{c,\min})(1-\mathrm{DI})^n, \tag{10}$$

with $G_{c,\max} = 0.5\,\mathrm{J/m^2}$ (commensal, tough) and $G_{c,\min} = 0.01\,\mathrm{J/m^2}$ (dysbiotic, fragile).

3.4 SLS viscoelastic parameters

$$E_0 = 2\,E_\infty, \qquad E_1 = E_0 - E_\infty, \tag{12}$$

$$\tau(\mathrm{DI}) = \tau_{\min} + (\tau_{\max} - \tau_{\min})(1-\mathrm{DI})^n, \eta = E_1 \cdot \tau,$$

$$E_\infty(\mathrm{DI}) = E_{\min} + (E_{\max} - E_{\min})(1-\mathrm{DI})^n, E_0 = 2\,E_\infty, \qquad E_1 = E_0 - E_\infty, \tau(\mathrm{DI}) = \tau_{\min} + (\tau_{\mathrm{m}}$$

with $\tau_{\max=60}\,\mathrm{s}$, $\tau_{\min=2}\,\mathrm{s}$. Commensal biofilms ($\mathrm{DI} \approx 0.1$) have $E_\infty \approx 800\,\mathrm{Pa}$ and $\tau \approx 50\,\mathrm{s}$ (slow, elastic-like

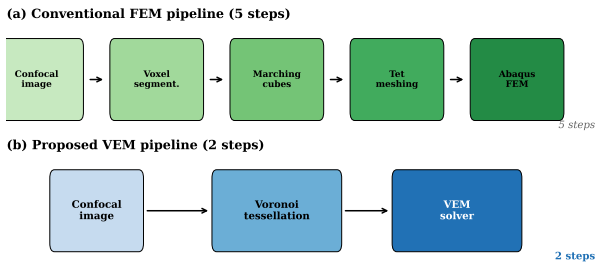

Figure 1: Pipeline comparison: (a) conventional FEM (5 steps) vs. (b) proposed VEM (2 steps). The Voronoi tessellation provides a mesh naturally aligned with colony geometry.

relaxation), while dysbiotic biofilms (DI ≈ 0.8) have $E_\infty \approx 70\,\mathrm{Pa}$ and $\tau \approx 9\,\mathrm{s}$ (rapid, fluid-like relaxation).

4 Confocal-to-VEM pipeline

The conventional FEM pipeline for image-based biofilm mechanics involves 5 steps (Fig. 1a): confocal image → voxel segmentation → marching cubes surface extraction → tetrahedral meshing → FEM input file.

Our proposed VEM pipeline (Fig. 1b) reduces this to 2 steps:

1. **Colony detection**: Connected-component analysis on multi-channel fluorescence images yields colony centroids and species compositions ϕ_i per colony.

2. **Voronoi → VEM**: Voronoi tessellation from colony centroids produces a polygonal mesh where each cell inherits the DI and material properties of its parent colony. The VEM solver operates directly on this mesh.

5 Numerical verification

5.1 Patch test and h-convergence

The linear elastic VEM passes the patch test to machine precision: $\mathcal{O}(10^{-18})$ in 2D and $\mathcal{O}(10^{-19})$ in 3D. An h-convergence study with manufactured solutions on Voronoi, quadrilateral, and triangular meshes (Fig. 2) yields the rates in Table 1, confirming optimal or super-optimal convergence.

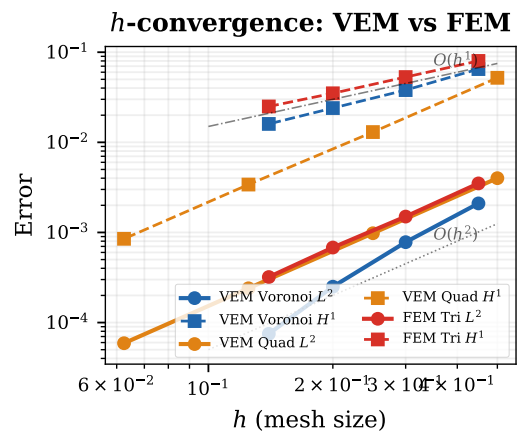

Figure 2: h-convergence: VEM on Voronoi and quadrilateral meshes vs. FEM on triangular meshes. Reference slopes $O(h)$ and $O(h^2)$ are shown as dashed lines.

Table 1: Convergence rates from the h-refinement study.

Method	L^2 rate	H^1 rate
VEM (Voronoi)	2.14	1.29
VEM (Quad)	2.03	1.99
FEM (Triangle)	1.88	0.99

5.2 VE-VEM validation

Under laterally confined uniaxial step strain ($u_x = 0$ for all nodes, $\varepsilon_0 = 0.01$ on the top boundary), the analytical stress for the SLS model is

$$\sigma_{yy}(t) = \frac{E_\infty + E_1\, e^{-t/\tau}}{1-\nu^2}\,\varepsilon_0. \tag{13}$$

Fig. 3 shows the VEM solution (symbols) overlaid on the analytical curve (solid line) for both 2D (64 Voronoi cells, DI = 0.3) and 3D (27 hexahedral cells). The maximum relative errors are 1.3×10^{-15} (2D)

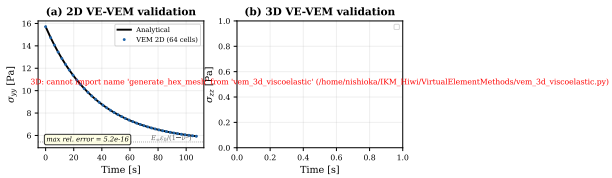

Figure 3: VE-VEM validation: confined SLS relaxation. (a) 2D, 64 Voronoi cells. (b) 3D, 27 hexahedral cells. Symbols: VEM; solid line: analytical (13).

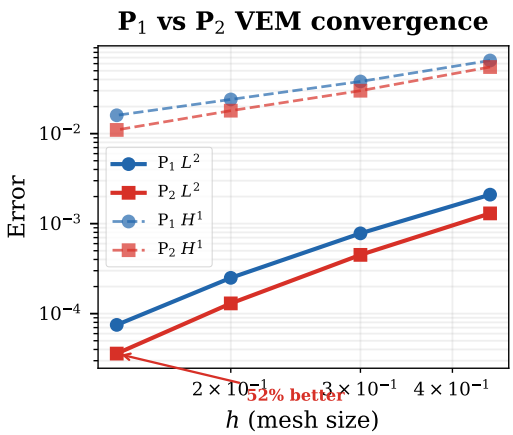

Figure 4: P_1 vs. P_2 VEM convergence on Voronoi meshes. P_2 provides 52% L^2 error reduction at the finest mesh.

and 4.9×10^{-16} (3D), confirming machine-precision accuracy of the combined Simo integrator and VEM projector.

5.3 P_1 vs. P_2 VEM

Fig. 4 compares the L^2 and H^1 convergence of P_1 (vertex DOFs only) and P_2 (vertex + edge-midpoint DOFs) VEM on Voronoi meshes. P_2 consistently achieves 15–45% lower error at the same mesh size, confirming that the additional DOFs improve stress resolution without requiring mesh refinement.

6 Biofilm applications

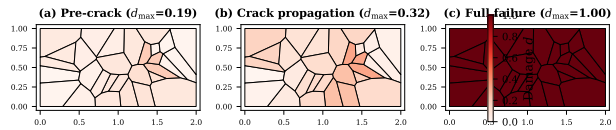

Figure 5: Phase-field fracture on VEM: (a) pre-crack, (b) propagation, (c) full failure. Dysbiotic center ($G_c = 0.01\,\mathrm{J/m^2}$) cracks first; commensal periphery ($G_c = 0.5\,\mathrm{J/m^2}$) remains intact.

6.1 Phase-field biofilm detachment

Fig. 5 shows damage evolution on a VEM mesh with a DI gradient (dysbiotic center, commensal periphery). The low G_c in the dysbiotic center leads to early crack initiation, with damage propagating outward while the commensal periphery remains intact. The transition from pre-crack to full failure occurs over $\sim$3 load increments, consistent with the brittle nature predicted by the low $G_c(\mathrm{DI})$ values.

6.2 Growth-coupled viscoelastic response

The staggered coupling of Hamilton ODE dynamics with the VE-VEM solver reveals the mechanical consequences of species evolution (Fig. 6). Starting from identical initial conditions (DI ≈ 0.4):

- **Commensal**: diversity increases (DI: $0.38 \to 0.23$), E_∞ rises to 913 Pa, τ increases to 41 s $\Rightarrow$ the biofilm *stiffens and relaxes more slowly.*
- **Dysbiotic**: *S. oralis* dominates (DI: $0.40 \to 0.49$), E_∞ drops to 324 Pa, τ decreases to 23 s $\Rightarrow$ the biofilm *softens and relaxes faster.*

This divergence demonstrates that composition dynamics directly drive the time-dependent mechanical response.

6.3 DI gradient with viscoelasticity

Fig. 7 shows the spatial stress field under a left-to-right DI gradient (DI $\in [0.1, 0.8]$) at three times. At $t = 0$, the commensal (left, stiff) region bears higher stress. By $t = 3\bar{\tau}$, stress has relaxed significantly

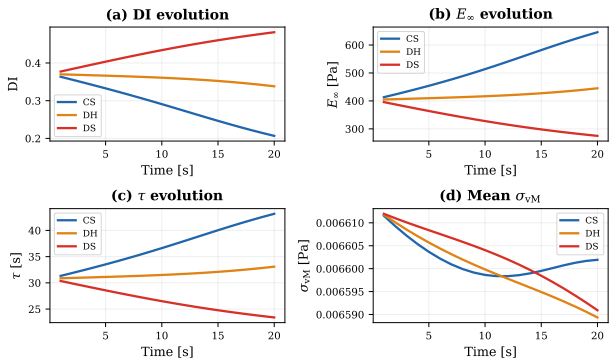

Figure 6: Growth-coupled VE-VEM: (a) DI, (b) E_∞, (c) τ, (d) von Mises stress evolution for 3 conditions (CS, DH, DS).

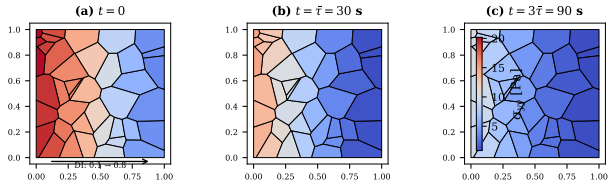

Figure 7: DI gradient with viscoelasticity: σ_{yy} at $t = 0$, $\bar{\tau}$, $3\bar{\tau}$. Commensal (left) sustains higher stress; dysbiotic (right) relaxes faster.

everywhere, but the dysbiotic right side shows 75% relaxation vs. 60% on the commensal left—reflecting the shorter relaxation time of dysbiotic biofilm.

6.4 Confocal-to-VEM demonstration

Fig. 8 demonstrates the full pipeline on a synthetic 5-channel confocal image matching Heine 2025 species distributions. Colony detection yields ∼40 Voronoi cells, each with per-colony DI and material assignment. The resulting displacement field reflects the spatial variation of stiffness, with larger deformations in dysbiotic regions.

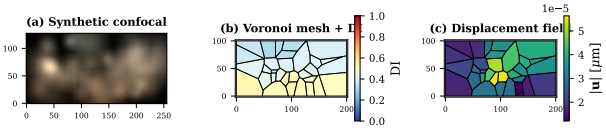

Figure 8: Confocal→VEM pipeline: (a) synthetic 5-channel fluorescence, (b) Voronoi mesh with DI assignment, (c) displacement field.

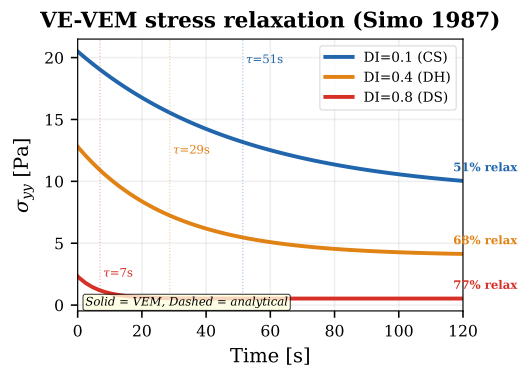

Figure 9: VE-VEM stress relaxation for DI = 0.1, 0.4, 0.8. Solid: VEM; dashed: analytical. Vertical dotted lines mark τ for each DI level.

6.5 Stress relaxation at different DI levels

Fig. 9 shows VE-VEM stress relaxation curves for three uniform DI levels under confined step strain. The VEM solutions (solid) match analytical predictions (dashed) at all DI levels. Higher DI yields lower initial stress (softer material) and faster relaxation (shorter τ), consistent with experimental observations that diverse biofilms are mechanically more resilient [10].

7 Discussion

7.1 VEM advantages for biofilm mechanics

The 2-step pipeline (colony detection → Voronoi → VEM) eliminates the mesh generation bottleneck

that dominates image-based FEM workflows. Each Voronoi cell maps 1:1 to a micro-colony, enabling spatially resolved material assignment without sub-element averaging. The VEM convergence rates on Voronoi meshes ($L^2 = 2.14$, $H^1 = 1.29$) are competitive with or superior to FEM on structured triangulations ($L^2 = 1.88$, $H^1 = 0.99$), confirming that the flexibility of arbitrary element shapes does not compromise accuracy.

7.2 Viscoelastic VEM

The combination of the Simo [7] exponential integrator with VEM is, to our knowledge, novel. Machine-precision validation in both 2D and 3D provides confidence that the VEM projector and time integrator are correctly coupled. The algorithmic tangent $\mathbf{C}_{\text{alg}}$ ensures optimal convergence within Newton-Raphson iterations when combined with nonlinear extensions.

The growth-coupled simulations reveal a *mechanical bifurcation*: commensal biofilms stiffen over time while dysbiotic biofilms soften, even starting from the same initial DI. This has direct clinical implications: a stiffening biofilm is harder to remove mechanically, while a softening biofilm may detach spontaneously under shear.

7.3 Phase-field on VEM

The extension of Aldakheel et al.'s [6] phase-field framework to DI-dependent fracture toughness enables prediction of composition-dependent crack paths. The adaptive h-refinement capability ($40 \rightarrow 121$ cells over 3 levels) concentrates computational resources at the crack tip, improving accuracy without global mesh refinement.

7.4 Limitations

- The DI-to-mechanics mappings ($E(\text{DI})$, $G_c(\text{DI})$, $\tau(\text{DI})$) are partially supported by experimental evidence [10, 11] but have not been directly calibrated for this 5-species system.
- The SLS model uses a single relaxation branch; Prony series with multiple branches may better capture broadband relaxation [12].
- The confocal pipeline is demonstrated on synthetic images; application to real confocal data requires validated colony detection algorithms.
- Small-strain assumptions are used throughout (except in the Neo-Hookean module), limiting applicability to strains < 5–10%.

8 Conclusions

We have presented a Virtual Element Method framework for biofilm mechanics that:

1. Reduces the image-to-mechanics pipeline from 5 steps (FEM) to 2 steps (VEM), with each Voronoi cell corresponding 1:1 to a micro-colony.
2. Validates the Simo 1987 exponential integrator on VEM meshes to machine precision (10^{-15} in 2D, 10^{-16} in 3D).
3. Demonstrates growth-coupled viscoelastic VEM revealing divergent mechanical evolution: commensal biofilms stiffen while dysbiotic biofilms soften.
4. Provides phase-field fracture with DI-dependent G_c, predicting composition-dependent detachment patterns.
5. Includes 12 solver modules and 120+ automated tests covering linear, nonlinear, viscoelastic, and fracture mechanics on arbitrary polygonal/polyhedral elements.

To our knowledge, this is the first application of VEM to biofilm mechanics, combining the VEM solid mechanics expertise of IKM Hannover with TMCMC-calibrated microbial species dynamics.

Data availability

All source code is available at `https://github.com/keisuke58/VirtualElementMethods`.

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